AudioApp Project Proposal

Professional Audiometry Mobile Application

Prepared for: Project Stakeholders
Prepared by: Moonarc Development Team
Date: January 16, 2026
Version: 1.0

Executive Summary

AudioApp is a professional-grade mobile audiometry solution that transforms hearing healthcare by enabling clinical-quality hearing assessments on consumer devices. The application serves three distinct user roles—patients, audiologists, and system administrators—providing both self-screening capabilities and diagnostic audiometry tools.

Value Proposition

Traditional Audiometry	AudioApp
Equipment cost: $3,000 - $15,000	Subscription-based model
Requires dedicated sound booth ($10,000+)	Any iOS/Android device
Fixed clinical location only	Portable + telehealth enabled
30+ minute setup	< 5 minute setup

Market Opportunity

- **466 million people** worldwide have disabling hearing loss
- **Global hearing aids market**: $9.2B (2024) → $15.8B (2030)
- **Mobile health market**: Projected to reach $150B by 2028
- **OTC hearing aid market**: $1B+ annually (new since 2022)

Project Scope

Core Features

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│                  AudioApp                      │
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│       Patient    │       ⚕ Audiologist  │       System Admin       │
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│ • Self-screening │ • Diagnostic   │ • User management         │
│ • View results   │   audiometry   │ • Calibration profiles    │
│ • Track hearing  │ • Bone conduct │ • Analytics dashboard     │
│ • Educational    │ • Reports/PDF  │ • Audiologist verification │
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Technical Specifications

Specification	Details
Platform	iOS (primary), Android (via Flutter)
Frequencies	250, 500, 1000, 2000, 3000, 4000, 6000, 8000 Hz
Intensity Range	-10 to +100 dB HL (5 dB increments)
Accuracy Target	±5 dB vs. clinical audiometer
Framework	Flutter 3.x + Dart 3.x
State Management	Riverpod
Database	SQLite (encrypted) + Firebase
Compliance	ANSI S3.6, ISO 8253, HIPAA, GDPR

Development Timeline & Phases

Phase 1: Core Architecture (Weeks 1-2)

- User role system and authentication
- Role-based routing
- Database schema migration
- Basic admin dashboard shell

Phase 2: Patient Screening Module (Weeks 3-4)

- Screening test interface
- Environment check system
- Pass/fail result screens
- Patient home screen

Phase 3: Admin Features (Weeks 5-6)

- User management screens
- Audiologist verification workflow
- Calibration profile management
- Basic analytics dashboard

Phase 4: Headphone Calibration & Validation Tools (Weeks 7-8)

- Headphone detection service
- Calibration profile database
- Validation test mode
- Deviation reporting

Phase 5: Clinical Validation (Weeks 9-12)

- Partner with audiologists for testing
- Run validation against reference audiometers
- Build validated headphone profile database
- Document accuracy statistics

Phase 6: Polish & Launch (Weeks 13-16)

- UI/UX refinements
- Tutorial and help system
- App Store submission preparation
- Marketing materials

Budget Breakdown

Development Costs

Category	Description	Estimated Cost
Flutter Development	Core app development (16 weeks)	$40,000 - $60,000
Audio Engine Development	Native iOS/Android audio with calibration	$15,000 - $25,000
UI/UX Design	Professional medical-grade interface	$8,000 - $12,000
Backend Infrastructure	Firebase setup, cloud services	$3,000 - $5,000
Testing & QA	Unit, integration, and device testing	$5,000 - $8,000
Development Subtotal		**$71,000 - $110,000**

Regulatory & Compliance Costs

Category	Description	Estimated Cost
Regulatory Consultant	FDA 510(k) or De Novo pathway guidance	$15,000 - $30,000
FDA Submission Fees	Pre-submission meeting + submission	$10,000 - $25,000
HIPAA Compliance Audit	Third-party security audit	$5,000 - $10,000
IEC 62304 Documentation	Software lifecycle documentation	$8,000 - $15,000
Regulatory Subtotal		**$38,000 - $80,000**

Clinical Validation Costs

Category	Description	Estimated Cost
Audiologist Partners	Clinical validation study (5-10 audiologists)	$10,000 - $20,000
Reference Equipment	Access to calibrated audiometers & sound booths	$5,000 - $10,000
Participant Compensation	60-75 participants @ $50-100 each	$3,000 - $7,500

IRB Approval	Institutional Review Board fees	$1,500 - $3,000
Study Report & Analysis	Statistical analysis and documentation	$3,000 - $5,000
Validation Subtotal		**$22,500 - $45,500**

Deployment & Operations

Category	Description	Estimated Cost
App Store Fees	Apple Developer + Google Play (annual)	$225/year
Cloud Services (Year 1)	Firebase, hosting, CDN	$3,000 - $6,000
Professional Insurance	Medical device liability insurance	$5,000 - $15,000/year
Marketing & Launch	App Store optimization, initial marketing	$5,000 - $10,000
Deployment Subtotal		**$13,225 - $31,225**

Total Project Investment

Scenario	Estimated Total
Minimum (Screening Only - Lower Regulatory)	**$100,000 - $130,000**
Standard (Full Features - 510(k) Pathway)	**$145,000 - $200,000**
Premium (Extended Validation + Enterprise)	**$200,000 - $270,000**

> [!NOTE]
> The "Screening Only" scenario focuses on general wellness claims with enforcement discretion, avoiding the full FDA 510(k) submission process. The "Standard" scenario includes full diagnostic features requiring regulatory approval.

Regulatory Pathway

Option A: General Wellness / Screening Only (Lower Cost)

Claims Allowed:

- "Check your hearing health"
- "Track changes in hearing over time"
- "Identify when to see a professional"
- "Diagnose hearing loss" (not allowed)

Regulatory Burden: Minimal
Cost Savings: ~$35,000 - $60,000

Option B: Class II Medical Device (510(k))

When Required:

- Making diagnostic claims
- Targeting healthcare professionals
- Claiming clinical accuracy

Timeline: 12-24 months for FDA approval

Phase	Duration
Pre-Submission Meeting	2-3 months
Analytical Testing	2-3 months
Clinical Validation	3-4 months
Submission Preparation	2-3 months
FDA Review	3-12 months

Revenue Model

Tier	Monthly Price	Target Users
Patient Free	$0	Self-screening, basic results
Patient Premium	$4.99	Detailed reports, history, tracking
Audiologist Solo	$49/month	Full diagnostics, 50 patients
Audiologist Clinic	$149/month	Unlimited patients, multi-user
Enterprise	Custom	API access, white-label, EMR integration

Projected Revenue (Year 1-3)

Metric	Year 1	Year 2	Year 3
Audiologist Subscribers	50	200	500
Patient Premium Subscribers	200	1,000	5,000
Monthly Recurring Revenue	$3,500	$15,000	$40,000
Annual Revenue	**$42,000**	**$180,000**	**$480,000**

Risk Assessment

Technical Risks

Risk	Probability	Impact	Mitigation
Audio accuracy insufficient	Medium	Critical	Early validation, audio engineer consultant
Device fragmentation	High	High	Extensive testing, calibration database
App Store rejection	Low	High	Engage Apple early, ensure compliance

Regulatory Risks

Risk	Probability	Impact	Mitigation
FDA requires full submission	Medium	High	Consult regulatory expert, budget accordingly
HIPAA violation	Low	Critical	Security audit, legal review
Liability concerns	Medium	High	Strong disclaimers, professional insurance

Market Risks

Risk	Probability	Impact	Mitigation
Low audiologist adoption	Medium	High	Beta program, testimonials, competitive pricing
Competition from established players	Medium	Medium	Focus on UX and mobile-first features

Team Requirements

Role	Status	Priority
Flutter Developer	In progress	-
Audiologist Advisor	Needed	High
Regulatory Consultant	Needed	High
Clinical Validation Partner	Needed	High
Audio Engineer (Consultant)	Optional	Medium

Success Criteria

Technical Metrics

- Audio accuracy: ±5 dB vs. clinical audiometer
- App crash rate: < 0.1%
- Test completion time: < 15 minutes average

Business Metrics (Year 1)

- 100 paid audiologist users
- User retention: > 60% at 6 months
- Customer satisfaction: > 4.5/5 rating

Clinical Metrics

- Test-retest reliability: r > 0.90
- Sensitivity for hearing loss detection: ≥ 90%
- Specificity for normal hearing: ≥ 80%

Recommended Next Steps

1. **Approve Budget Tier** - Decide between Screening Only, Standard, or Premium pathway
2. **Engage Regulatory Consultant** - Critical for FDA pathway decision
3. **Secure Audiologist Partners** - Needed for clinical validation
4. **Begin Phase 1 Development** - Core architecture and authentication
5. **Schedule FDA Pre-Submission Meeting** - If pursuing 510(k) pathway

Appendix: Current Development Status

Component	Status	Notes
Technical Specification	Complete	Full 954-line spec document
Architecture Design	Complete	Clean layer separation
FDA Compliance Guide	Complete	All pathways documented
Clinical Validation Protocol	Complete	IRB-ready
Implementation Plan	Complete	8-week core development

Core App Structure	In Progress	Flutter framework established
Audio Engine	In Progress	Basic tone generation working
Testing Screens	In Progress	Manual audiometry functional
Patient Screening	Not Started	Awaiting approval
Admin Dashboard	In Progress	Basic structure complete
Clinical Validation	Not Started	Requires audiologist partners

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Contact & Approval

Project Lead: Moonarc Development Team
Stakeholder Approval Required By: [DATE]

Approval	Name	Signature	Date
Budget Approval			
Regulatory Pathway			
Go/No-Go Decision			

This proposal is confidential and intended for internal stakeholder review only.